

Using the Kano Method to Evaluate Features of Chatbot-Based Technological Support for Recovery from Alcohol Use Disorder

Yu-Ching Lin
National Tsing Hua University
Hsinchu, Taiwan
yumi90203@gapp.nthu.edu.tw

Yan-Ming Chen
National Tsing Hua University
Hsinchu, Taiwan
gepz@gapp.nthu.edu.tw

Chih-Ching Chuang
National Tsing Hua University
Hsinchu, Taiwan
ch3@gapp.nthu.edu.tw

Chien Wen (Tina) Yuan
D-School
National Taiwan University
Taipei City, Taiwan
tinayuan@ntu.edu.tw

Nanyi Bi
Department of Information
Management
National Taiwan University
Taipei, Taiwan
nanyibi@ntu.edu.tw

Hu-Ming Chang
Taipei City Hospital Songde Branch
Taipei, Taiwan
dau66@tpech.gov.tw

Ming-Chyi Huang
Department of Psychiatry, Taipei City
Psychiatric Center
Taipei City Hospital Songde Branch
Taipei, Taiwan
mch@tpech.gov.tw

Chuang-Wen You
Graduate Institute of Art and
Technology
National Tsing Hua University
Hsinchu, Taiwan
cwyou@mx.nthu.edu.tw

Abstract

Chatbots are emerging as a promising tool for the provision of continuous care outside clinical settings; however, the perspectives of both patients and their companions on engaging with chatbot-based support for AUD interventions remain largely underexplored. This study employed a Kano two-factor model to evaluate the integration of chatbot-based technological support in AUD treatment. Our analysis revealed that patients tend to focus on self-management, while their companions prioritize communication and emotional support. Patients were enthusiastic about adjusting chatbot personality settings according to their current emotional state. They also strongly favored having the chatbot update their companions and act as a proxy in providing encouragement. These insights underscore the importance of designing chatbot systems tailored to both patients and companions, particularly in the development of personality settings and emotional support features.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**.

Keywords

Alcohol Use Disorders, Kano Method, Chatbot Technology.

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1 Introduction

Alcohol Use Disorder (AUD) refers to the compulsive use of alcohol despite the risk of harm to the individual's physical, mental, and social well-being [39, 40]. However, the effectiveness of outpatient programs [51] in monitoring patients is often hindered by the infrequency of interactions. Smartphones provide access to chatbot technology from anywhere at any time [37], making them a valuable tool for maintaining continuous engagement with patients between appointments. This has prompted many app developers to prioritize chatbots as a key channel for user engagement.

Chatbot systems have proven highly effective in providing mental health support for conditions, such as dementia [47], stress [26], and severe depression [16]. The conversational interface of chatbots is particularly conducive to managing the emotional disorders with which many alcohol-dependent patients must contend. Nonetheless, most previous studies have evaluated chatbots solely in terms of medical outcomes, largely overlooking exploring individuals' willingness to engage with specific features that these systems can offer.

In this survey study, the Kano two-factor model [6] was used to evaluate the integration of Chatbot technology designed to facilitate recovery from alcohol dependence. Our primary objective was to understand how patients and trusted companions (e.g., family members) perceive these innovations and derive insights to guide future development. This analysis revealed important differences

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between patients and their companions in classifying features as One-dimensional or Attractive (explained later in Section 3.3). Three key insights emerged from this analysis: (1) Patients prioritize self-management, whereas companions prioritize communication and emotional support; (2) Emotional support is a key element in chatbot interactions dealing with AUD; and (3) Chatbot personality settings can have a powerful influence on the effectiveness of emotional support mechanisms and the willingness of users to participate.

2 Background & Related Work

The Kano model is widely used in product design and development, market research, and quality management to classify and prioritize customer needs according to their impact on customer satisfaction [6, 45]. This system categorizes customer-requested features into six categories [6]. *Must-be* features (M): Basic expectations. Fulfillment does not increase satisfaction (as it is assumed), and absence causes dissatisfaction. *One-dimensional* features (O): Features explicitly desired by users. Fulfillment ensures satisfaction, and absence causes dissatisfaction. *Attractive* features (A): Features that exceed expectations. Fulfillment enhances satisfaction, but their absence does not cause dissatisfaction. *Indifferent* features (I): Features with little to no impact on user satisfaction whether present or absent. *Reverse* features (R): Features that decrease satisfaction when included and increase satisfaction when omitted. *Questionable* features (Q): Features with unclear or contradictory feedback, making it difficult to assess their impact on satisfaction [6]. Even before features are fully developed, various studies [19, 41, 54] have analyzed user needs through simplified simulations or concept validation to ensure alignment with user expectations.

In our literature review, we first summarize chatbot features relevant to AUD treatment. Next, we provide examples of potential chatbot techniques to enhance user experience. Finally, we identify social sensing solutions that could be effectively integrated with chatbot technology, as detailed below.

2.1 Existing Chatbot Solutions for AUD

Jovanović et al. [24] classified conversational healthcare chatbots according to three primary service archetypes: (1) chatbots for diagnosis, (2) chatbots for prevention, and (3) chatbots for therapy, which are summarized as follows.

First, chatbots for diagnosis have demonstrated moderate accuracy in suggesting diagnoses based on case vignettes, even at their early stage of development [25]. These chatbots enhance user engagement with mental health assessments, facilitating the identification of individuals in need of care [20]. Machine learning approaches analyzing behavioral patterns and self-reported data show potential in predicting the onset of mental health conditions [30]. Additionally, integrating chatbots with established tools like the AUDIT questionnaire [5] could predict drinking episodes by analyzing behavioral and self-reported data [27].

Second, chatbots for prevention build on the progress made in self-monitoring mobile apps for alcohol use management [8, 13, 14, 52, 53]. The recent emergence of chatbot technology offers new possibilities, as studies indicate their potential as valuable tools for individuals with substance use disorders [36].

Lastly, chatbots for therapy have been explored in recent research for their application in treating substance use disorders. Studies highlight their ability to provide real-time, tailored support to patients, underscoring their promise in addressing addiction challenges [9, 38, 50].

2.2 Enhancing User Experience with Chatbot Humanization

Chatbots are conversational AI applications designed to enhance the user experience by mimicking human-to-human interactions through text, voice, or a combination of the two [32]. These chatbot systems facilitate interactions that are natural and engaging [42] while providing flexibility by allowing users to re-read and revise their responses [34].

Additionally, chatbot configuration can be used to influence the appearance of a chatbot as well as its persona, communication style [11], and the sophistication of its conversational intelligence [22]. Chatbots can be personalized and foster trust through anthropomorphism [48], linguistic style adaptation [2], and prompt engineering [49]. These capabilities have the potential to support AUD patients and their companions in daily life but require input from users to identify the features they find most valuable and necessary.

2.3 Social Sensing in Support Channels

Social sensing technology enables the sharing of behavioral data with concerned significant others (CSOs), helping them to support patients in maintaining abstinence [17, 18, 46]. In addiction-related healthcare, there has been a shift away from restrictive mediation (imposing restrictions) in favor of active mediation (collaborating with patients), or a combination of the two, fostering a more meaningful and supportive recovery process. Mobile apps have been developed to facilitate communication between patients with alcohol addiction and their caregivers or family members [52]. However, the emotional support provided by these systems is often quite basic, relying on one-time responses rather than the dynamic, interactive communication that chatbots can provide.

Advanced AI systems like ChatGPT help organize and share information between patients and caregivers, creating dynamic, personalized interactions tailored to users' emotional states [23]. This technology can also be used to formulate communication strategies and strengthen emotional connections between patients and support personnel, serving as potential design features tailored to the needs of AUD patients and their companions.

Our aim in this study was to apply the Kano model to identify the needs of alcohol-dependent patients and their support companions throughout the recovery process. Feature assessment was based on the following research questions:

- *RQ1*: What features could be included in a chatbot-based support system to make it easier for patients and their companions to deal with the process of recovering from alcohol use disorder?
- *RQ2*: How do patients and their companions perceive these features in terms of Kano categories and satisfaction indices, and what are the implications for the design of support systems?

- *RQ3*: How do the needs and desires of patients differ from those of their companions, and what are the implications for the design of support systems?

3 Survey Study based on the Kano two-factor method

The primary objective of this survey study was to evaluate innovations in chatbot technology from the perspectives of alcohol-dependent patients and their companions.

3.1 Procedure

Patients and their support companions were recruited through hospital referrals. Eligible patients were identified in the offices of two collaborating psychiatrists (co-authors of this paper). Screening was based on the following inclusion criteria: 1) aged 18 to 70 years, 2) meeting DSM-5 criteria for Alcohol Use Disorder (AUD) [4], 3) currently receiving outpatient treatment for AUD, 4) no dependence on other substances, 5) no diagnosis of serious mental illnesses (e.g., schizophrenia or bipolar disorder), and 6) the ability to understand and provide informed consent. Most of the patients visited the hospital alone; however, a small number were accompanied by companions (e.g., family members). Both patients and companions were asked whether they would be willing to participate in a survey. Some companions declined, due to time constraints or a reluctance to disclose their association with alcohol dependence. After explaining the research goals, the study participants were shown 15 design features using screenshots and functional descriptions of existing chatbot applications [8, 10, 13, 44]. To facilitate a more intuitive understanding, the various features were presented as practical scenarios within a chat application similar to Facebook Messenger, with the chatbot acting as the opposite party. Insights gained from this process informed the design of features for a chatbot-based support system (outlined in Section 3.3). Throughout this presentation, emphasis was placed on ensuring that participants understood how the features (e.g., sharing behavioral data or exchanging messages) could facilitate the recovery process. Participants (patients and companions) were then guided through a Kano questionnaire assessing features designed for deployment on smartphones. This preliminary assessment was meant to collect valuable user insights early in the product development process, helping to refine feature designs and reduce development costs.

3.2 Participants

This study recruited 30 patients (P01-P30) from Taipei City Hospital, ranging in age from 30 to 65 years ($M = 46.92$, $SD = 8.83$). Ten patients (P01–P10) participated in the survey with one companion each. The companions (C01–C08) were either the patients' partner, parent, or sibling, ranging in age from 39 to 63 years. Roughly one-third of the participants (9 patients and 4 companions) had prior experience using chatbot technologies (e.g., Siri or chatbots for banking or healthcare services) in their daily lives. After the specifics of the chatbot technology had been outlined, nearly all participants (29 patients and all 10 companions) expressed a willingness to use a chatbot to help manage AUD.

3.3 Kano Survey Questionnaire

Chatbot features identified in a review of the literature were categorized into five primary groups: (A) Diagnosing Alcohol Use Disorder (AUD), (B) Maintaining abstinence/ preventing relapse, and (C) Providing therapeutic support. We also formulated two novel features for assessment, including (D) Personalized chatbot experiences and (E) Incorporating social sensing, as detailed below. Our analysis indicated that by querying patients about AUD symptoms [43] and providing a database of diagnosis-related information [1], it would be possible to create a chatbot capable of assisting in diagnosis and generating recommendations for relevant medical services. See Features 1 and 2 (Diagnosing Alcohol Use Disorder).

Patients with AUD typically undergo initial treatment to facilitate withdrawal, followed by maintenance treatment to prevent relapse. During the withdrawal phase, patients are closely monitored by therapists; however, during the maintenance phase, they are personally responsible for adhering to medication and psychiatrist instructions. Some patients forget to take their medication, necessitating reminders [31]. Other patients have chronic conditions (e.g., high blood pressure), necessitating an understanding of potential medication interactions [12, 15]. Thus, we posited using use chatbot technology to provide personalized reminders to take medication and guidance in the use of medications. See Feature 3 (Maintaining abstinence and preventing relapse).

Maintaining abstinence depends on the patient's ability to self-monitor their recovery progress, as well as guidance from friends and family in adopting a healthy lifestyle. Researchers have noted the benefits of patients working with companions to set recovery goals and ensure those goals are fulfilled [7]. Proactive monitoring updates have also proven effective in detecting abnormalities in self-monitoring. Theoretically, chatbot technology could be utilized to track patient's progress by visualizing self-monitoring data, negotiating recovery goals, and providing proactive updates to keep all stakeholders informed. See Features 4–6 (Maintaining abstinence and preventing relapse).

Companionship makes it easier for many patients to resist the temptation of alcohol. Thus, we proposed the creation of a chatbot that engages with patients proactively by querying their behavior and emotional states in accordance with the principles of Cognitive Behavioral Therapy (CBT) [44]. The sense of humanity could be enhanced by emulating human-like interactions, such as interest-driven messages, jokes, inspirational quotes, or content from the 12-step program. The chatbot could also be programmed to reach out to patients on a regular schedule or after periods of inactivity. Expressions of empathy and the use of uplifting language would foster a supportive environment, promote a positive mindset, and enhance connectedness. Suggestions for engaging activities and topics of interest could move the patient beyond a myopic focus on their struggles with alcoholism. See Features 7–9 (Providing therapeutic support).

User engagement could also be enhanced by personalizing the chatbot experience. As outlined in Section 2.2, developers could use prompts to customize the personality of the chatbot and filter out words with potentially negative connotations. The chatbot could also facilitate communication between patients and companions by enabling voice inputs for those who prefer not to type and allowing the revision or deletion of messages after they are sent. See Features 10–12 (Personalizing the chatbot experience).

As outlined in Section 2.3, the chatbot can generate strategies to facilitate communication between patients and companions, such as reminding companions not to blame the patient for a lack of willpower, and emphasizing that alcohol dependence is a chronic brain disorder, with relapse not being viewed as a complete failure. It can also provide suggestions to overcome communication barriers and real-time updates, reminding companions to offer support. See Features 13–15 (Incorporating social sensing).

3.4 Measures

The proposed features were divided into six categories in accordance with the Kano model, as follows: Must-be features (M), Performance features (P), Attractive features (A), Indifferent features (I), Reverse features (R), and Questionable features (Q). The categories were then ranked according to impact on user satisfaction, as follows: Must-be > One-dimensional > Attractive > Indifferent > Reverse [6]. Questionable features (Q) were excluded to prevent misinterpretation by participants or ambiguity in the questionnaire design.

The CS^+ and CS^- parameters [6] were used to assess the degree to which the features contributed to basic satisfaction, where $CS^- < -0.50$ indicates a feature's contribution to basic dissatisfaction, and $CS^+ > 0.50$ reflects a feature's potential to create positive satisfaction. By prioritizing high CS^+ scores, we derived criteria to guide the development of products capable of meeting user needs and ensuring user satisfaction.

3.5 Data analysis

Python libraries were used to analyze data related to the 15 design features, analyzing the entire sample as well as the patient and companion groups separately. In cases where the Fong test indicated a significant difference, mode statistics [6] was used to classify features into the appropriate Kano model category. In cases where the difference did not meet the significance threshold, we followed the decision rule outlined in [6] for feature categorization. We also calculated the coefficient of satisfaction (CS^+) and the coefficient of dissatisfaction (CS^-) to highlight key design features. Features where $CS^- < -0.50$ had a greater impact on customer dissatisfaction when excluded, while those with $CS^+ > 0.50$ had a higher effect on customer satisfaction when included [33].

4 Results

4.1 Feature Categories: One-dimensional and Attractive

As shown in the Total Sample column in Table 1, three features were categorized as One-dimensional. All three were deemed moderately attractive ($CS^+ > .50$). The data from companions for these three features align with the total sample data for two of the features, while one feature shows higher attractiveness. This indicates that for these features, the expectations and satisfaction levels of companions were similar to those in the Total sample. Our analysis of patient data revealed several discrepancies within the group. For example, the feature Sharing through discreet chatbot update was classified as Indifferent by patients ($CS^+ = 0.31$), indicating that it was perceived as largely unimportant.

Further analysis of the total sample data revealed 6 features in the Attractive category. All six features were consistently classified as Exciting in both patient and companion data, indicating that

both groups considered these features to have a positive impact on user satisfaction. All features in the Maintaining and preventing relapse group were regarded as extremely important in both data sets, emphasizing their roles in promoting the recovery process.

Finally, we analyzed data from the two groups separately. When assessed by patients, nine features were categorized as Attractive or One-dimensional. One feature, Visualizing self-monitoring data, was excluded from this analysis as it was specifically designed for patients. Most of the remaining 8 features aligned with the companions' needs, indicating that both parties assessed Attractive or One-dimensional features in a similar manner.

4.2 Variability in Feature Categorization: Patients versus Companions

Notable discrepancies were observed between patients and companions in classifying features as Indifferent (six features marked in green in Table 1). When assessed by patients, these six features were viewed as Indifferent, with satisfaction coefficients (CS^+) below 0.50 and dissatisfaction coefficients (CS^-) of no lower than -0.50. When assessed by companions, these six features fell within the Attractive or One-dimensional categories. This indicates that patients were only weakly in favor of these features. In contrast, companions classified the same features as either Attractive or One-dimensional, reflecting a stronger preference.

In the companion data, four of the features earned a CS^+ score equal to or greater than 0.70, while two earned scores of 0.50. This indicates that companions had high expectations for most of these features, suggesting a slight decrease in satisfaction if the feature were absent.

These findings highlight notable differences between patients and companions in the prioritization of functional needs, particularly those associated with communication and social sensing. Patients appeared indifferent to features related to social sensing, which focuses on understanding interpersonal interactions or social behaviors. Further exploration is needed to understand the reasons behind this indifference and to identify design enhancements that could bridge this gap. These insights could inform future chatbot design and feature prioritization to better address the needs of both patients and companions.

5 Discussion

5.1 Revisiting the Results: Patients Emphasize Self-Management and Companions emphasize Communication and Emotional Support

Patients categorized three features as one-dimensional, while the companions assigned four to this category. Two of these features were categorized similarly by both groups ("Partnering in CBT interactions" and "Enabling voice communication with chatbot"). The former is crucial to identifying and understanding negative emotions, thoughts, and behavioral patterns, making it an indispensable component of chatbot interventions. The latter provides a convenient and natural approach to communication by eliminating the communication barriers inherent to text inputs.

Table 1: Categorical classification of the 15 features based on the CS^+ and CS^- . Abbreviations: C (Category), A (Attractive), O (One-dimensional), M (Must-be), I (Indifferent). “*” denotes a significant value based on the Fong test. Dark grey bars represent CS^+ (or CS^-) values exceeding 0.5 (or -0.5). Dark green highlights discrepancies in classification between patients and companions.

Feature	Total sample				Patient				Companion			
	C	CS ⁻	CS ⁺		C	CS ⁻	CS ⁺		C	CS ⁻	CS ⁺	
Diagnosing Alcohol Use Disorder (AUD)												
1. Querying patients about symptoms	I	-0.31		0.38	I	-0.33		0.33	A*	-0.20		0.50
2. Directing the patient toward medical services	I	-0.33		0.45	O*	-0.40		0.43	A*	-0.10		0.50
Maintaining and preventing relapse												
3. Providing personalized medication reminders	A	-0.38		0.72	A*	-0.47		0.67	A	-0.10		0.90
4. Chatbot visualizing self-monitoring data	A	-0.42		0.66	A*	-0.37		0.67	--	--		--
5. Negotiating recovery goals with chatbot	A*	-0.33		0.52	A*	-0.40		0.50	A*	-0.10		0.60
6. Proactive self-monitoring updates	A*	-0.34		0.54	A*	-0.34		0.55	A*	-0.30		0.50
Providing therapeutic support to assist therapy												
7. Partnering in CBT interactions	O*	-0.35		0.50	O*	-0.50		0.50	O*	-0.40		0.60
8. Offering activities or topics of interest	A*	-0.18		0.52	A*	-0.24		0.50	A*	-0.30		0.70
9. Proactive messages to express care	I*	-0.26		0.46	I*	-0.24		0.45	A*	-0.30		0.70
Personalized chatbot experience												
10. Customizing the chatbot personality	A*	-0.25		0.52	A*	-0.27		0.50	A*	-0.20		0.60
11. Customizing a word filter	I	-0.28		0.46	I	-0.27		0.43	A*	-0.30		0.50
12. Enabling voice communication with chatbot	O*	-0.48		0.52	O*	-0.43		0.50	O*	-0.60		0.60
Incorporating social sensing												
13. Generating tips to enhance communication	I	-0.29		0.41	I	-0.31		0.45	O*	-0.60		0.70
14. Sharing through discreet chatbot updates	O*	-0.38		0.52	I	-0.24		0.31	A*	-0.40		0.70
15. Sharing companions' encouragement	I	-0.31		0.49	I	-0.28		0.38	O*	-0.40		0.80

The feature “Directing the patient toward medical services” presented divergent categorization. Patients classified it as One-dimensional (additive and necessary), whereas companions categorized it as Attractive (additive but unnecessary/extra). This feature aims to help patients overcome the lack of knowledge about medical resources or reluctance to seek medical help due to past experiences, making it easier for them to find trustworthy medical institutions, understand clinic hours, and learn how to make appointments. As an attractive feature, it also gives companions peace of mind, alleviating concerns about treatment delays.

Companions categorized “Generating tips to enhance communication” and “Sharing encouragement” as One-dimensional, whereas patients assigned them to the Indifferent category. This means that companions view these features as valuable – Helping them better understand how to support the patient, reduce conflict risks during interactions, and promote smooth communication. This indifference may stem from patients’ resistance to interaction with companions during the alcohol cessation process or a lack of motivation to actively improve communication. Moreover, Patients may be indifferent to encouragement from companions due to desensitization caused by long-term alcohol use. [28].

Six attractive features received consistent evaluations from both patient and companion groups. Features in the “Maintaining and preventing relapse” group were seen as an unexpected feature with practical value in the recovery process. For example, “Providing personalized medication reminders” and “Chatbot visualizing self-monitoring data” can reduce the risk of relapse, help patients intuitively understand their progress, and generate motivation [19]. Companions viewed the Supervision of daily health management

as a valuable feature, capable of alleviating their sense of helplessness in the face of volatile emotions and behaviors [3]. Overall, the features deemed Attractive by both sides were those with practical value in supporting treatment and recovery.

5.2 Using Emotional Support Functions to Enhance Chatbot Interactions

A comparison of patient and companion data revealed six features with clear discrepancies between the two groups (highlighted in green in Table 1). Most of these features were categorized as Attractive or One-dimensional by companions and can be found in the “Incorporating social sensing” group, which addresses functions related to human interactions and social behavior. Companions generally view the process of communicating with the patient as the most important aspect, and expect the chatbot to provide communication strategies to deal with the patient’s emotional fluctuations and needs.

Four of these features received very high CS^+ ratings ($CS^+ > 0.50$) for companions: “Proactive messages to express care”, “Generating tips to enhance communication”, “Sharing through discreet chatbot updates”, and “Sharing companions’ encouragement.” These features highlight the importance of communication in the caregiving process. Two of the features received moderately high ratings ($CS^+ = 0.50$): “Querying patient about symptoms” and “Customizing a word filter.” The moderately high score may indicate that family communication tends to be more direct, slightly reducing the need for medical advice or word filtering. Our results showed that

companions expect the chatbot to facilitate bidirectional communication to cope with emotional fluctuations, while patients focus more on medical advice.

Families affected by alcohol abuse are prone to emotional stress when faced with relapse, often leading to hostile language and belligerent behavior, further exacerbating frustration, a negative interaction environment, and/or avoidance patterns [21]. Patients are unable or unwilling to engage in constructive dialogue, especially while drinking, making it difficult to address emotional issues within the family [29]. Patients often develop self-protective mechanisms to avoid confronting their emotional state [35]. This is evident in their indifference toward features designed to facilitate communication and emotional processing.

This clear divergence of opinions related to these essential issues highlights the difficulties faced by developers in balancing the needs of both companions and patients in the design of emotional support functions. This may involve prioritizing the needs of one party or adopting a flexible approach capable of meeting the needs of both parties.

5.3 Chatbot Personalization and Activity Suggestions to Enhance Emotional Support

The “Customize the Chatbot Personality” feature received very high CS^+ ratings ($CS^+ > 0.50$) among companions, likely due to its provision of emotional support. Many companions working with individuals struggling with alcohol dependence often face psychological challenges such as depression and helplessness. For patients, however, the feature received only moderately high ratings ($CS^+ = 0.50$). The numbing effect of alcohol on emotions can make it difficult for patients to recognize and express their feelings, which in turn impacts their interpersonal relationships [55]. This reduced emotional responsiveness may explain why the feature achieved lower scores among patients compared to the higher ratings observed among companions.

This may explain the enthusiasm patients expressed about tailoring the tone of the chatbot (warm, rational, or humorous) according to their immediate needs. If the chatbot is set to act as a friend, conversations will be relaxed and lighthearted; if set as a healthcare professional, it will adopt a more rigid approach to implementing the regimen. Adjusting the chatbot’s role according to the patient’s needs can effectively enhance emotional support and improve its approachability and reliability, helping patients and companions establish long-lasting interactions [55].

6 Conclusion

This study employing the Kano two-factor model to explore the use of chatbots to support Alcohol Use Disorder (AUD) treatment. Findings revealed that patients prioritize self-management, while companions focus on communication and emotional support. Patients valued adjusting chatbot personality settings to match their emotional state and favored the chatbot acting as a proxy to update and encourage companions. These insights highlight the need for chatbot designs tailored to both patients and companions, emphasizing personalized settings and emotional support features.

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